DA3451+DA3452 — Computer Vision / Speech & Language Technology (Core Basket I)

Tier	B
Phase (order)	5
Do this first	DA3409
Estimated effort	12 weeks each
Unlocks	DA6800, DA7016

Roadmap

1. CV: CNN detection and segmentation, Szeliski.
2. NLP: tokenisation, n-grams, attention, transformers, Jurafsky & Martin.

Video courses (all links checked on YouTube on 2026-10-22)

🎓 Main course: follow it start to end

- Stanford CS231N Deep Learning for Computer Vision I 2025 (18 videos). Stanford CS231N *Deep Learning for Computer Vision* (2025, official). Pick this if you take the CV option.
- Stanford CS224N Natural Language Processing with Deep Learning I Spring 2024 I Professor Christopher Manning (23 videos). Stanford CS224N *NLP with Deep Learning* (official). Pick this if you take the speech & language option.

🎥 Other full lecture series (pick one if the main one doesn't click, or for topics it skips)

- Speech Technology_Prof. Umesh (73 videos). IIT Madras BS *Speech Technology* (S. Umesh): the speech half (DA3452).
- Deep Learning for Computer Vision (22 videos). Michigan EECS 498 *Deep Learning for Computer Vision* (Justin Johnson).

Study plan (week by week)

Lecture numbers are positions in the main playlist above (the first one). Click a number to open that video.

Week	Lectures	What to do
1	1–3	Image classification, linear classifiers, regularisation, optimisation. Assignment 1 (kNN, SVM, softmax) from cs231n.github.io.
2	4–6	Backprop, CNNs, CNN architectures (ResNet etc.). Assignment 2: CNN on CIFAR-10.

Week	Lectures	What to do
3	7–8	RNNs, attention, transformers.
4	9–10	Detection, segmentation, visualisation, video. Fine-tune a YOLO / Mask R-CNN model.
5	11–14	Distributed training, self-supervised learning, generative models (VAE, GAN, diffusion).
6	15–18	3D vision, vision-language, robot learning. Speech option: follow Prof. Umesh's Speech Technology series (alt) instead of this plan.

▼ **All lectures in *Stanford CS231N Deep Learning for Computer Vision I 2025* (18 videos)**

1. Stanford CS231N Deep Learning for Computer Vision | Spring 2025 | Lecture 1: Introduction
2. Stanford CS231N | Spring 2025 | Lecture 2: Image Classification with Linear Classifiers
3. Stanford CS231N | Spring 2025 | Lecture 3: Regularization and Optimization
4. Stanford CS231N | Spring 2025 | Lecture 4: Neural Networks and Backpropagation
5. Stanford CS231N | Spring 2025 | Lecture 5: Image Classification with CNNs
6. Stanford CS231N Deep Learning for Computer Vision | Spring 2025 | Lecture 6: CNN Architectures
7. Stanford CS231N | Spring 2025 | Lecture 7: Recurrent Neural Networks
8. Stanford CS231N | Spring 2025 | Lecture 8: Attention and Transformers
9. Stanford CS231N | Spring 2025 | Lecture 9: Object Detection, Image Segmentation, Visualizing
10. Stanford CS231N Deep Learning for Computer Vision | Spring 2025 | Lecture 10: Video Understanding
11. Stanford CS231N | Spring 2025 | Lecture 11: Large Scale Distributed Training
12. Stanford CS231N | Spring 2025 | Lecture 12: Self-Supervised Learning
13. Stanford CS231N Deep Learning for Computer Vision | Spring 2025 | Lecture 13: Generative Models 1
14. Stanford CS231N Deep Learning for Computer Vision | Spring 2025 | Lecture 14: Generative Models 2
15. Stanford CS231N Deep Learning for Computer Vision | Spring 2025 | Lecture 15: 3D Vision
16. Stanford CS231N Deep Learning for Computer Vision | Spring 2025 | Lecture 16: Vision and Language
17. Stanford CS231N Deep Learning for Computer Vision | Spring 2025 | Lecture 17: Robot Learning
18. Stanford CS231N Deep Learning for Computer Vision | Spring 2025 | Lecture 18: Human-Centered AI

▼ **All lectures in *Stanford CS224N Natural Language Processing with Deep Learning I Spring 2024 I Professor Christopher Manning* (23 videos)**

1. Stanford CS224N: NLP with Deep Learning | Spring 2024 | Lecture 1 - Intro and Word Vectors
2. Stanford CS224N: NLP with Deep Learning | Spring 2024 | Lecture 2 - Word Vectors and Language Models
3. Stanford CS224N: NLP with Deep Learning | Spring 2024 | Lecture 3 - Backpropagation, Neural Network
4. Stanford CS224N: NLP with Deep Learning | Spring 2024 | Lecture 4 - Dependency Parsing
5. Stanford CS224N: NLP with Deep Learning | Spring 2024 | Lecture 5 - Recurrent Neural Networks
6. Stanford CS224N: NLP with Deep Learning | Spring 2024 | Lecture 6 - Sequence to Sequence Models
7. Stanford CS224N: NLP w/ DL | Spring 2024 | Lecture 7 - Attention, Final Projects and LLM Intro
8. Stanford CS224N NLP with Deep Learning | 2023 | Lecture 8 - Self-Attention and Transformers

9. Stanford CS224N NLP with Deep Learning | 2023 | Lecture 9 - Pretraining
10. Stanford CS224N NLP with Deep Learning | 2023 | Lecture 11 - Natural Language Generation
11. Stanford CS224N: NLP with Deep Learning | Spring 2024 | Lecture 10 - Post-training by Archit Sharma
12. Stanford CS224N: NLP with Deep Learning | Spring 2024 | Lecture 11 - Benchmarking by Yann Dubois
13. Stanford CS224N: NLP w/ DL | Spring 2024 | Lecture 12 - Efficient Training, Shikhar Murty
14. Stanford CS224N: NLP w/ DL| Spring 2024 | Lecture 13 - Brain-Computer Interfaces, Chaofei Fan
15. Stanford CS224N: NLP w/ DL | Spring 2024 | Lecture 14 - Reasoning and Agents by Shikhar Murty
16. Stanford CS224N: NLP with Deep Learning | Spring 2024 | Lecture 15 - After DPO by Nathan Lambert
17. Stanford CS224N: NLP with Deep Learning | Spring 2024 | Lecture 16 - ConvNets and TreeRNNs
18. Stanford CS224N: NLP with Deep Learning | Spring 2024 | Lecture 18 - NLP, Linguistics, Philosophy
19. Stanford CS224N NLP with Deep Learning | 2023 | Lecture 16 - Multimodal Deep Learning, Douwe Kiela
20. Stanford CS224N NLP with Deep Learning | 2023 | Lec. 19 - Model Interpretability & Editing, Been Kim
21. Stanford CS224N NLP with Deep Learning | 2023 | Python Tutorial, Manasi Sharma
22. Stanford CS224N NLP with Deep Learning | 2023 | PyTorch Tutorial, Drew Kaul
23. Stanford CS224N NLP with Deep Learning | 2023 | Hugging Face Tutorial, Eric Frankel

Books, notes and websites

- Szeliski (free); Stanford CS231n
- Jurafsky & Martin SLP3 (free); Stanford CS224n; Hugging Face courses

Practice / proof of understanding

Train a detector or fine-tune a transformer for text; write a short report on it.

What your IITM course sheet says

From the PDF: DA3451

Official description: Objectives: The course provides the student a solid foundation in the principles and techniques of computer vision with emphasis on deep learning based techniques

Official syllabus (trimmed):

> Course contents: 1. Introduction to Computer Vision and Deep Learning ? Overview of computer vision and deep learning ? Basic Image Processing ? History and development of deep learning in computer vision ? Basic principles of artificial neural networks 2. Convolutional Neural Networks (CNNs) - Review 3. Object Detection and Localization ? Faster R-CNN, YOLO, SSD 4. Image Segmentation ? Understanding image segmentation ? Semantic segmentation with Fully Convolutional Networks (FCN) ? Instance segmentation with Mask R-CNN, semantic segmentation with Encoder-Decoder Architecture, U-Nets 5. Generative Adversarial Networks (GANs) ? Understanding GANs and their architecture ? Training GANs for image generation and manipulation ? Applications of GANs in computer vision - Super-Resolution 6. Diffusion-Based Generative Models - Applications- Text to Image- Captioning 7. Advanced Topics ? Motion Estimation - Optical Flow ? Structure From Motion ? Depth Estimation ? 3D Reconstruction ? Rendering - Neural Radiance Fields ? Medical image analysis with deep learning - Image Registration, Image Segmentation

Official textbooks: 1. Szeliski, Richard. Computer vision: algorithms and applications. Springer Nature, 2022.

Official references: Reference books: 1. Forsyth, D.A. and Ponce, J., 2002. Computer vision: a modern approach. prentice hall 2. Hartley, Richard, and Andrew Zisserman. Multiple view geometry in computer vision. Cambridge university press, 2003

From the PDF: DA3452

Official description: : This course introduces the computational methods used to process human language and speech. The course covers the representation and processing of text and speech signals, statistical and neural language modeling, and modern architectures such as transformers and large language models. Students will learn how machines process language and speech through techniques including tokenization, probabilistic language models, neural networks, and deep learning based models for speech recognition and synthesis.

Official syllabus (trimmed):

> Introduction to Speech and Language Technologies: Overview of speech and language technologies and their applications. Historical development of the field and interdisciplinary foundations. Applications such as search engines, conversational agents, translation systems, and voice assistants. Words and Tokens: Representation of text in computational systems. Tokenization and normalization. Subword tokenization methods. Edit distance algorithms and their applications. N-gram Language Models: Probabilistic language modeling and the chain rule of probability. Unigram, bigram, and higher order n gram models. Training and evaluation using perplexity. Word Embeddings: Distributed representations of words and semantic similarity. Learning embeddings from large corpora. Word2Vec models including CBOW and Skip Gram. GloVe embeddings and applications of word vectors in downstream tasks. Neural Networks for Language Processing: Neural language models. Representation learning for language. Comparison with traditional language modeling approaches. Transformers: Self attention mechanisms for sequence modeling. Encoder and decoder architectures. Multi head attention and positional encodings. Transformers as the foundation of modern language models. Large Language Models: Pretraining large neural language models using large scale corpora. Generative language modeling objectives.

Official textbooks: 1. Daniel Jurafsky and James H. Martin. Speech and Language Processing. 3rd Edition. Stanford University

Official references: 1. Lewis Tunstall, Leandro von Werra, Thomas Wolf. Natural Language Processing with Transformers. O’Reilly Media. 2022.